TECHNICAL TESTING REPORT

Title :

Group Name :
Technical Evaluators (IT Experts) _____

Instructions :

1)This report is to be done and tested by the programmers and system analysts of the group and 10 IT experts .

2) Fill out the details before starting the testing procedure . LIST ALL THE MODULES / FUNCTIONALITY OF YOUR SYSTEM, DELETE THOSE WHO ARE NOT INCLUDED IN YOUR SYSTEM. (YOU CAN ALSO GET THEM FROM YOUR MIDTERM PROGRESS REPORT IF AVAILABLE)
3. Record and document the testing specially when there are bugs or errors.
4) Functional Testing / Test Case or / Testing Plan (copy paste when you are done to your documentation Testing and Procedures)
5) When the testing is completed , errors are fixed then you can proceed to SYSTEM EVALUATION.

COMPREHENSIVE SOFTWARE TECHNICAL TESTING REPORT

1. Document Information

Item	Details
Project/System Name	_____

Project Title	_____
Version	_____
Testing Date	_____
Testing Environment	_____
Tested By	_____
Developer/Development Team	_____
Test Report Version	_____
Status	☐ Draft ☐ Final

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2. Executive Summary

2.1 Purpose of Testing

This technical testing report presents the procedures, test cases, results, identified defects, corrective actions, and overall assessment of the software system. The testing activities were conducted to determine whether the system meets its specified functional, technical, performance, security, usability, compatibility, and reliability requirements.

2.2 Testing Objectives

The testing objectives are to:

1. Verify that all major system functions operate according to specifications.
2. Identify defects, errors, and unexpected system behavior.
3. Evaluate system performance under defined conditions.
4. Verify compatibility across supported devices, browsers, and operating environments.
5. Assess system security and access-control mechanisms.
6. Verify database operations and data integrity.
7. Evaluate the system against applicable software quality standards.
8. Confirm that identified defects have been corrected and retested.
9. Determine whether the software is ready for deployment or user evaluation.

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3. System Overview

3.1 System Description

System Name: _____

Description:

_____

_____

_____

3.2 Intended Users

User Type	Description	Access Level
Administrator	_____	_____
Faculty/Staff	_____	_____

Student/User _____ _____

Other _____ _____

3.3 Major System Modules

No.	Module/Feature	Description
1	User Authentication	_____
2	Dashboard	_____
3	User Management	_____
4	Main Application Module	_____
5	Reports	_____
6	Database Management	_____
7	Other	_____

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4. Testing Scope

4.1 In-Scope

The following components were included in testing:

- User registration and authentication
- Login/logout
- User account management
- Core application functions
- Database operations
- Input validation
- Error handling
- Navigation
- API integration
- Security mechanisms
- Performance
- Browser/device compatibility
- Data storage and retrieval
- Reports and outputs

4.2 Out-of-Scope

The following were excluded from the current testing cycle:

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5. Testing Methodology

The testing process followed a structured testing approach consisting of:

Requirements → Test Planning → Test Case Development → Test Execution → Defect Identification → Correction → Retesting → Final Evaluation

Testing Types

Testing Type	Purpose	Performed?
Unit Testing	Test individual components	☐
Integration Testing	Test interaction between modules	☐
System Testing	Test the complete system	☐
Functional Testing	Verify required functions	☐
Usability Testing	Evaluate ease of use	☐
Performance Testing	Evaluate response and processing	☐
Security Testing	Identify security vulnerabilities	☐
Compatibility Testing	Test supported environments	☐
Database Testing	Verify data operations and integrity	☐
API Testing	Verify API requests/responses	☐
Regression Testing	Verify fixes did not introduce new defects	☐
User Acceptance Testing	Verify user requirements	☐

6. Test Environment

6.1 Hardware Environment

Component	Specification
Processor	_____
RAM	_____
Storage	_____
Display Resolution	_____
Network	_____
Other	_____

6.2 Software Environment

Software	Version
Operating System	_____
Web Browser	_____

Programming Language	_____
Framework	_____
Database	_____
Web Server	_____
API/LLM Service	_____
Other Dependencies	_____

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7. Test Data

Describe the data used during testing.

Data Category	Description	Number of Records
User Accounts	_____	_____
Sample Transactions	_____	_____
Test Questions/Content	_____	_____

Database Records	_____	_____
Other	_____	_____

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8. Functional Test Cases

Test Case Format

Field	Details
Test Case ID	TC-F-001
Module	_____
Test Objective	_____
Precondition	_____
Test Input	_____
Test Steps	_____
Expected Result	_____
Actual Result	_____

Status ☐ Pass ☐ Fail

Remarks _____

Functional Testing Summary

Test Case ID	Module	Test Description	Expected Result	Actual Result	Status
TC-F-001	Login	Valid credentials	User successfully logs in	_____	Pass/Fail
TC-F-002	Login	Invalid password	Error message displayed	_____	Pass/Fail
TC-F-003	Registration	Valid registration	Account created	_____	Pass/Fail
TC-F-004	Dashboard	Load dashboard	Dashboard displayed	_____	Pass/Fail
TC-F-005	Database	Save record	Record stored successfully	_____	Pass/Fail
TC-F-006	Search	Valid search	Correct results displayed	_____	Pass/Fail

9. Input Validation Testing

Test ID	Input Field	Test Condition	Expected Behavior	Actual Behavior	Status
IV-001	Username	Empty	Validation message	_____	Pass/Fail
IV-002	Email	Invalid format	Reject input	_____	Pass/Fail
IV-003	Password	Insufficient length	Reject input	_____	Pass/Fail
IV-004	Numeric Field	Text input	Reject invalid data	_____	Pass/Fail
IV-005	Required Field	Blank	Display required message	_____	Pass/Fail

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10. Database Testing

10.1 Database Operations

Test ID	Operation	Test	Expected Result	Actual Result	Status
DB-001	INSERT	Add record	Record successfully saved	_____	Pass/Fail

DB-002	SELECT	Retrieve record	Correct record returned	_____	Pass/Fail
DB-003	UPDATE	Modify record	Record updated	_____	Pass/Fail
DB-004	DELETE	Delete record	Record removed	_____	Pass/Fail
DB-005	Validation	Duplicate data	Duplicate prevented	_____	Pass/Fail

10.2 Data Integrity

Verify:

- ☐ Primary key integrity
- ☐ Foreign key integrity
- ☐ Referential integrity
- ☐ Required fields
- ☐ Data type consistency
- ☐ Duplicate prevention
- ☐ Transaction consistency
- ☐ Backup/recovery

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11. API / External Service Testing

For systems using APIs, cloud services, AI/LLM services, or third-party services:

Test ID	API/Service	Request	Expected Response	Actual Response	Status
API-001	_____	_____	_____	_____	Pass/Fail
API-002	_____	_____	_____	_____	Pass/Fail
API-003	_____	_____	_____	_____	Pass/Fail

API Evaluation

- ☐ Successful requests
- ☐ Invalid requests handled
- ☐ Authentication verified
- ☐ Timeout handling
- ☐ Error handling
- ☐ Response validation
- ☐ Rate-limit handling
- ☐ Service failure handling

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12. Performance Testing

Test ID	Performance Metric	Test Condition	Expected	Actual	Status
PT-001	Page Load Time	Normal load	≤ ___ sec	___ sec	Pass/Fail

PT-002	API Response	Normal request	≤ ___ sec	___ sec	Pass/Fail
PT-003	Database Query	Normal query	≤ ___ sec	___ sec	Pass/Fail
PT-004	Concurrent Users	___ users	Stable	_____	Pass/Fail
PT-005	Stress Test	High load	No critical failure	_____	Pass/Fail

Performance Metrics

- Average response time: _____
- Minimum response time: _____
- Maximum response time: _____
- Number of concurrent users: _____
- Error rate: _____
- System availability: _____

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13. Security Testing

Test ID	Security Area	Test Procedure	Expected Result	Actual Result	Status
SEC-001	Authentication	Invalid login	Access denied	_____	Pass/Fail

SEC-002	Authorization	Unauthorized page access	Access denied	_____	Pass/Fail
SEC-003	Password	Password storage	Securely handled	_____	Pass/Fail
SEC-004	Session	Logout	Session terminated	_____	Pass/Fail
SEC-005	Input	Malicious input	Properly handled	_____	Pass/Fail
SEC-006	API	Unauthorized API request	Request rejected	_____	Pass/Fail

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14. Compatibility Testing

Browser Compatibility

Browser	Version	Result	Remarks
Google Chrome	_____	Pass/Fail	_____
Microsoft Edge	_____	Pass/Fail	_____
Mozilla Firefox	_____	Pass/Fail	_____

Safari	_____	Pass/Fail	_____

Device Compatibility

Device	OS	Screen Size	Result
Desktop	Windows	_____	Pass/Fail
Laptop	Windows	_____	Pass/Fail
Tablet	Android/iOS	_____	Pass/Fail
Smartphone	Android/iOS	_____	Pass/Fail

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15. Usability and User Interface Testing

Test ID	Usability Criterion	Expected Result	Actual Result	Status
UI-001	Navigation	Easy to navigate	_____	Pass/Fail
UI-002	Readability	Text is readable	_____	Pass/Fail
UI-003	Consistency	UI elements consistent	_____	Pass/Fail

UI-004	Feedback	System provides feedback	_____	Pass/Fail
UI-005	Error Messages	Clear error messages	_____	Pass/Fail
UI-006	Responsiveness	UI adapts to screen	_____	Pass/Fail

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16. ISO/IEC 25010 Software Quality Evaluation

If the technical testing is aligned with **ISO/IEC 25010**, the following characteristics can be included:

Quality Characteristic	Evaluation Focus	Result
Functional Suitability	Completeness, correctness, appropriateness	_____
Performance Efficiency	Time behavior, resource utilization, capacity	_____
Compatibility	Co-existence and interoperability	_____
Interaction Capability	Ease and effectiveness of user interaction	_____
Reliability	Availability, fault tolerance, recoverability	_____
Security	Confidentiality, integrity, authentication, accountability	_____

Maintainability	Modularity, reusability, analysability, modifiability, testability	_____
Flexibility	Adaptability, scalability, installability, replaceability	_____
Safety	Risk identification and mitigation, where applicable	_____

Note: The exact ISO/IEC 25010 edition and terminology used should be stated in the methodology, because the quality model has been revised across editions.

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17. Defect / Bug Report

Defect ID	Module	Description	Severity	Priority	Status
BUG-001	_____	_____	Critical/High/Medium/Low	High/Med/Low	Open
BUG-002	_____	_____	Critical/High/Medium/Low	High/Med/Low	Fixed
BUG-003	_____	_____	Critical/High/Medium/Low	High/Med/Low	Retest

Severity Classification

Severity	Description
Critical	Prevents the system from operating or causes major data/security failure
High	Major function does not work correctly
Medium	Function works with limitations or workaround
Low	Minor defect with limited impact

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18. Corrective Action and Retesting

Defect ID	Corrective Action	Date Fixed	Retest Result	Final Status
BUG-001	_____	_____	Pass/Fail	Closed/Open
BUG-002	_____	_____	Pass/Fail	Closed/Open
BUG-003	_____	_____	Pass/Fail	Closed/Open

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19. Regression Testing

Regression testing was conducted after corrections and system modifications to determine whether previously functioning features remained operational.

Test ID	Previously Passed Function	Retest Result	Status
REG-001	Login	_____	Pass/Fail
REG-002	Registration	_____	Pass/Fail
REG-003	Database Operations	_____	Pass/Fail
REG-004	Main Feature	_____	Pass/Fail
REG-005	Reports	_____	Pass/Fail

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20. Testing Results Summary

Testing Category	Total Tests	Passed	Failed	Pass Rate
Functional	___	___	___	___%
Database	___	___	___	___%
API	___	___	___	___%

Performance	___	___	___	___%
Security	___	___	___	___%
Compatibility	___	___	___	___%
Usability	___	___	___	___%
Regression	___	___	___	___%
TOTAL	___	___	___	___**%**

Pass Rate Formula

Pass Rate = (Number of Passed Test Cases ÷ Total Test Cases) × 100

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21. Overall Technical Assessment

Overall Testing Result

☐ **Passed – Ready for Deployment**

☐ **Passed with Minor Issues**

☐ **For Further Improvement**

☐ **Failed – Major Corrections Required**

Technical Assessment

Based on the testing activities conducted, the software system:

The identified defects and issues were:

The system is therefore assessed as:

☐ **Technically Acceptable**
☐ **Acceptable with Minor Revisions**
☐ **Requires Further Testing**
☐ **Not Acceptable**

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22. Recommendations

Based on the testing results, the following recommendations are proposed:

1.
2.
3.
4.
5.

23. Testing Approval

Role	Name	Signature	Date
Test Engineer/Tester	_____	_____	_____
Developer	_____	_____	_____
Project Leader	_____	_____	_____
Technical Adviser	_____	_____	_____
Project Adviser	_____	_____	_____

24. Appendices

Appendix A – Test Case Documentation

Attach complete test cases.

Appendix B – Screenshots of Testing

Include screenshots showing:

- Successful test execution

- Failed test cases
- Error messages
- Corrective actions
- Retesting results

Appendix C – System Logs

Include relevant:

- Application logs
- API logs
- Error logs
- Database logs
- Performance logs

Appendix D – Test Environment

Provide detailed hardware and software specifications.

Appendix E – Defect Documentation

Attach detailed bug reports and corrective-action records.